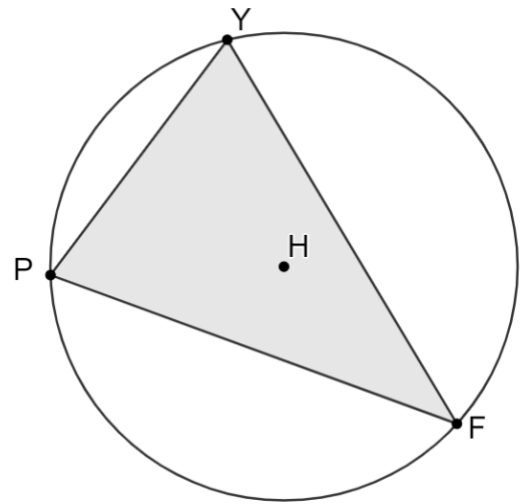


1. Circle H is shown where points Y , F , and P are on the circle. $m\widehat{FY} = (11n + 14)^\circ$, $m\angle PYF = (5n + 8)^\circ$, and $m\angle YFP = (3n + 3)^\circ$.
- a. Determine the value of n .



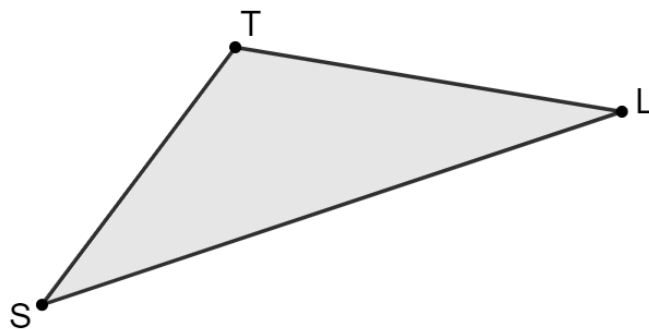
$n =$	12
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- b. Determine $m\widehat{PY}$.

$m\widehat{PY} =$	78°
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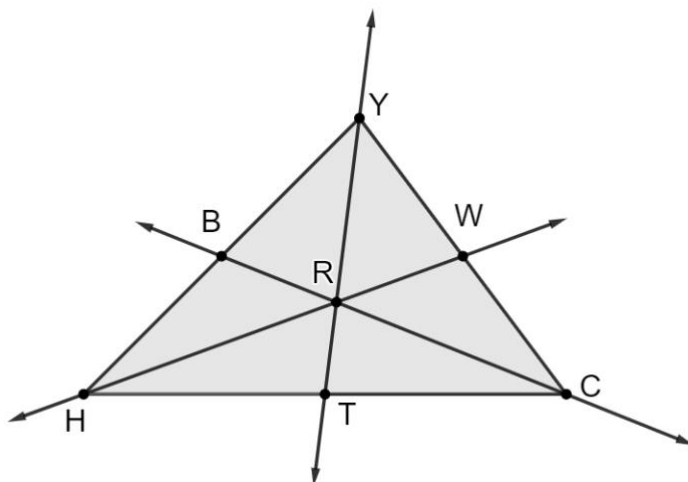
2. Triangle TLS is shown where $TL = \frac{92}{15}$, $TS = 5$, and $SL = \frac{237}{25}$. List the angles in order from least to greatest.

Least		Greatest
$\angle L$	$\angle S$	$\angle T$



3. Triangle YCH is shown where point R is the centroid. $RT = 13.4$, $RH = 39$, $YW = 25$, and $RB = 17.95$.

a. Determine the measure of $\overline{RC}$.

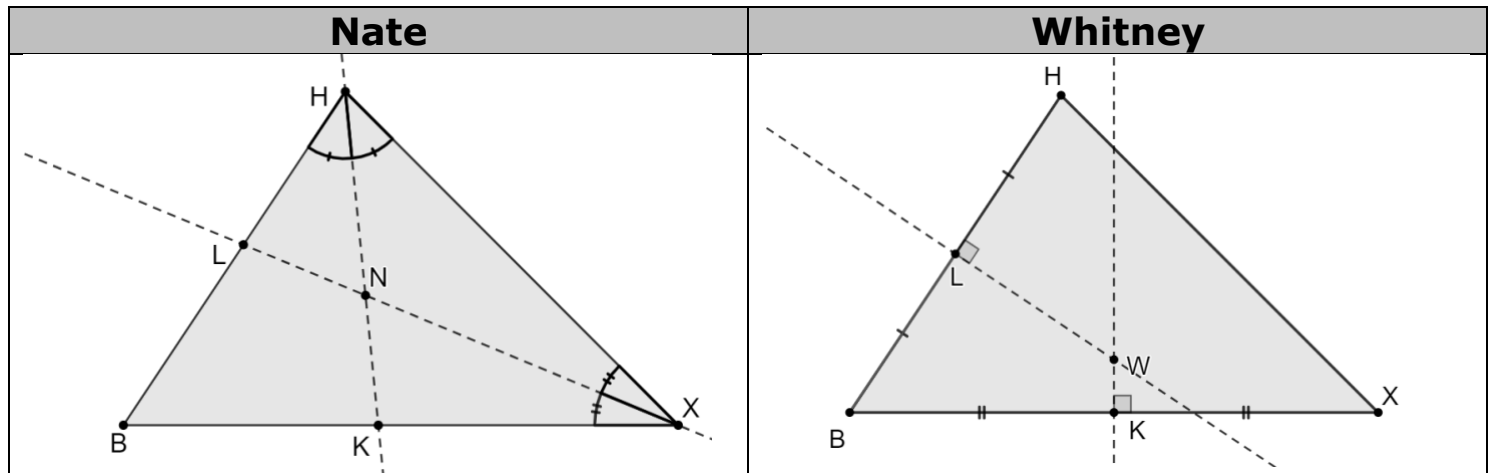


$RC =$	35.9
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b. Determine the perimeter of $\triangle YRW$.

Perimeter	71.3
-----------	------

4. Nate and Whitney used geometry software for two different constructions. Their incomplete constructions are shown.



Complete the statements.

☐ Nate's

☐ Whitney's

construction uses

☐ medians

☐ angle bisectors

☐ segment bisectors

☐ perpendicular bisectors

to construct

the incenter of $\triangle HBX$. When completed,

☐ Nate's

☐ Whitney's

construction will

result in a circle that is circumscribed about $\triangle HBX$ with

☐ $\overline{BN}$

☐ $\overline{BW}$

☐ $\overline{LN}$

☐ $\overline{LW}$

as a radius.

☐ Nate's

☐ Whitney's

construction can be used to show the Triangle

Sum theorem because the angles of $\triangle HBX$ are

☐ central angles

☐ inscribed angles

☐ angles formed by tangents

of the

☐ inscribed

☐ circumscribed

circle such that the three arcs intercepted by the

interior angles of $\triangle HBX$ complete a circle, so the sum of the measures of the interior angles of $\triangle HBX$ is half the measure of a circle.

5. Circle M is shown where points R, W, H, X are on the circle.

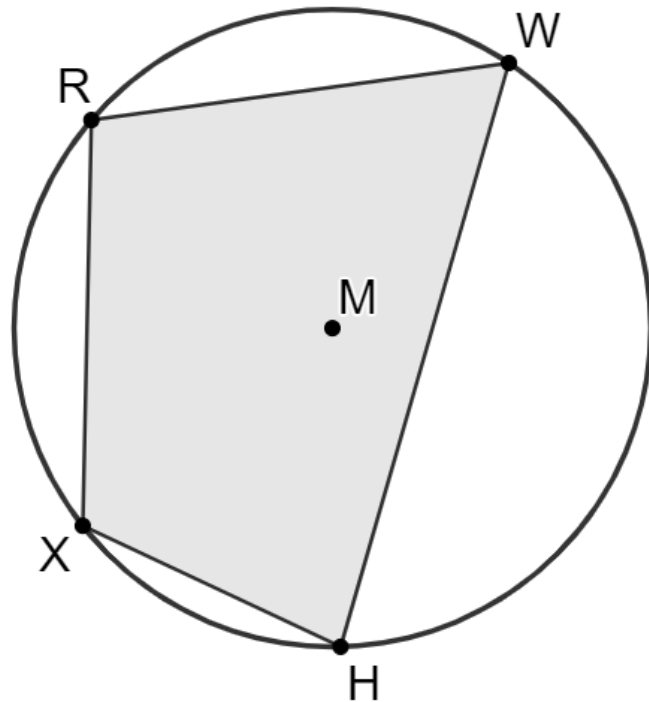
$$m\angle WRX = (5x + 14)^\circ,$$

$$m\angle RWH = (3x + 3y)^\circ,$$

$$m\angle RXH = (2x + 16y)^\circ, \text{ and}$$

$$m\angle WHX = (2x + 47)^\circ.$$

- a. Determine the value of x .



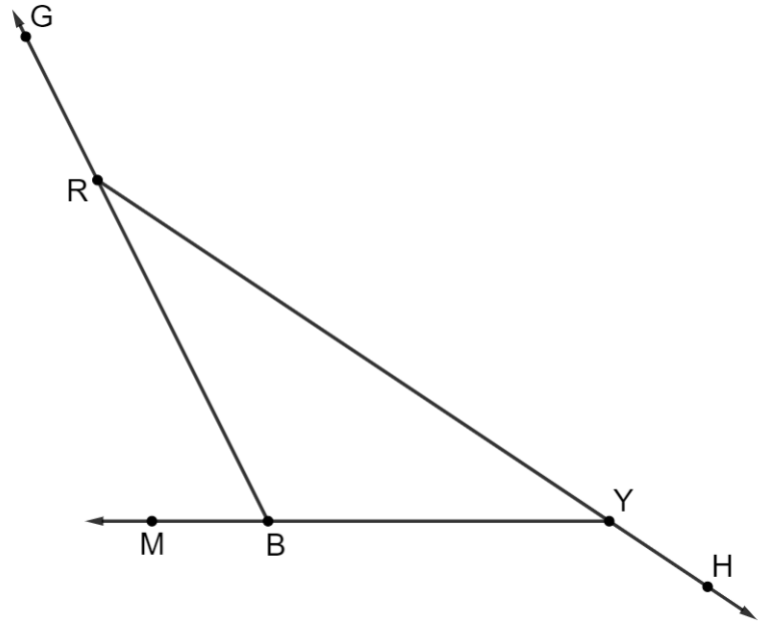
$x =$	17
-------	----

- b. Determine the value of y .

$y =$	5
-------	---

6. Triangle RYM is shown where $m\angle HYB = (8d + 9.5)^\circ$, $m\angle BRY = 29.7^\circ$, and $m\angle MBR = (4d - 5)^\circ$.

Determine the value of d .



$d =$	17.1
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7. Select all of the side lengths that can be used to create a triangle.

☒ 14, 14, 14

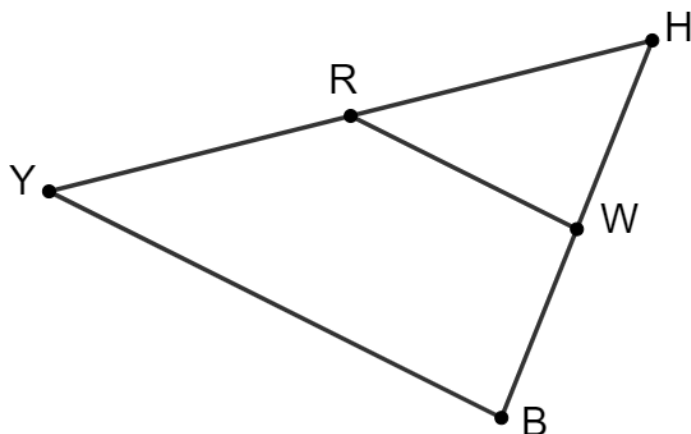
☐ 17, 18, 35

☐ $\frac{37}{4}, \frac{37}{8}, \frac{37}{12}$

☐ 23.1, 45.6, 73.4

☒ 45.62, 65.76, 111.21

Triangle YHB is shown where $\overline{RW}$ is a midsegment. $YB = 40.25$, $BH = 32.31$, and $YR = 24.74$.



8. Determine the measure of each.

	Segment	Length
a.	$\overline{RW}$	20.125
b.	$\overline{RH}$	24.74

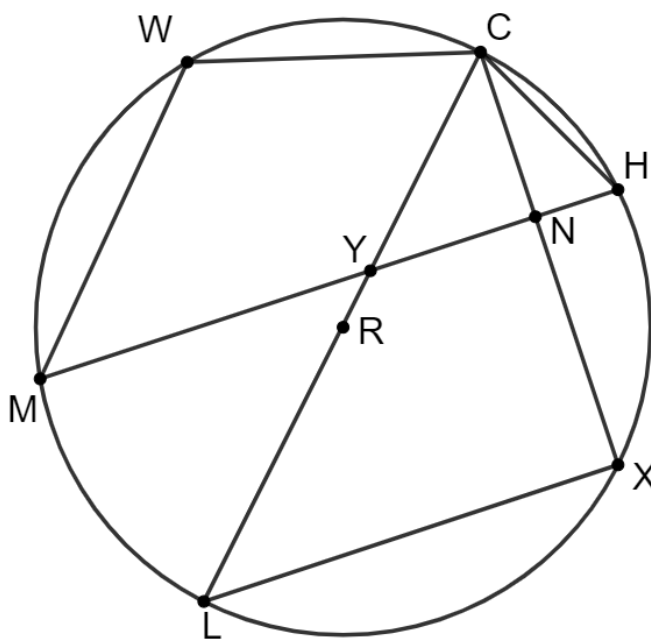
9. Select all of the statements that are true about $\triangle YHB$.

- ☐ $\overline{YH} \cong \overline{HB}$
- ☐ $\overline{YR} \cong \overline{BW}$
- ☒ $\triangle RHW \sim \triangle YHB$
- ☐ $\angle HRW \cong \angle HWR$
- ☒ $\angle HWR \cong \angle WBY$

10. Circle R is shown where points C , H , X , L , M , and W are on the circle.

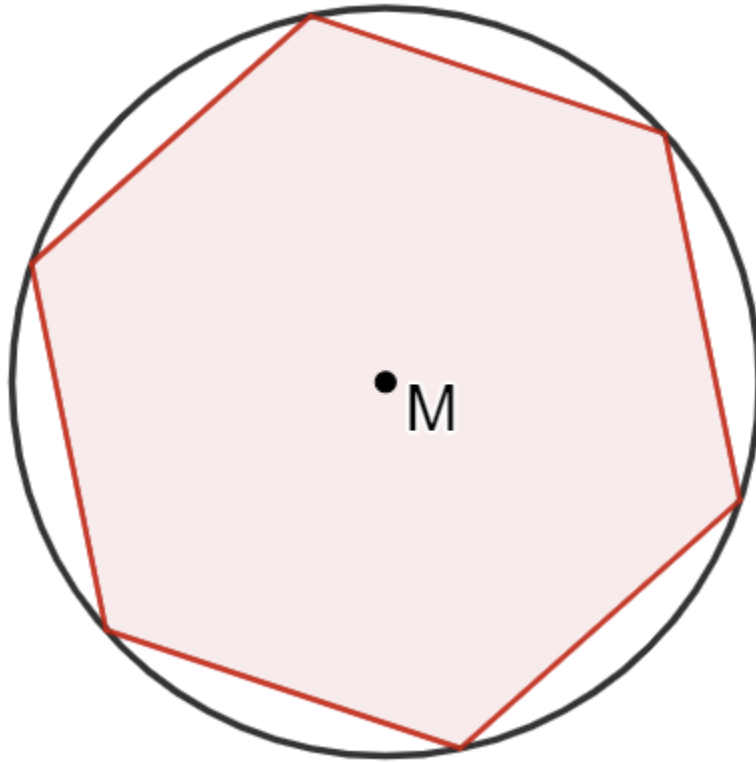
Select all of the statements that must be true.

- ☒ $m\angle LXC = 90^\circ$
- ☐ $m\angle CNH = 90^\circ$
- ☐ $m\angle XNY + m\angle YLX = 180^\circ$
- ☒ $m\angle MWC + m\angle MHC = 180^\circ$
- ☐ $m\angle MWC + m\angle WMH = 180^\circ$



11. Circle M is shown.

Answers may vary.



Use the circle to construct a hexagon that is inscribed in circle M .